# 3D-Printed Thermoplastic Polyurethane Housings for Force-Sensing Resistors in Biomedical Research

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## Motivation

- The foot insole will use center of pressure calculations to then give haptic feedback for balance with a vibrotactile belt.
- The scoliosis brace sensors are a clinician driven effort to introduce contact force quantification, as a better way to validate brace effectiveness.

## Methods and Design

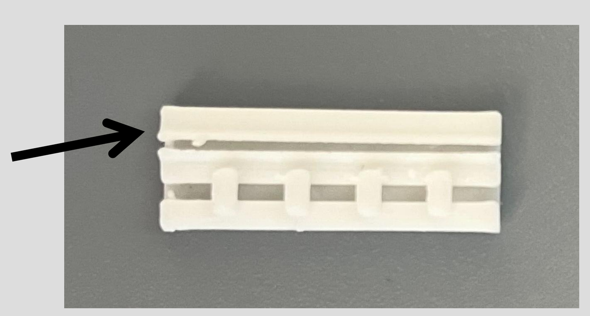
| Design Criteria | Specification |
|---|---|
| Customizability | Shoe Range (Men's): 4+ |
| Serviceability | Need to be able to access the wires and FSRs |
| Ease of Manufacturing | <1 hour (not including 3D printing) |
| Comfort | Flat (no wires sticking up), Flexible (Shore ≤95A) |
| Ease of Use | Plug-and-Play, Simple plug in |

Design Process:

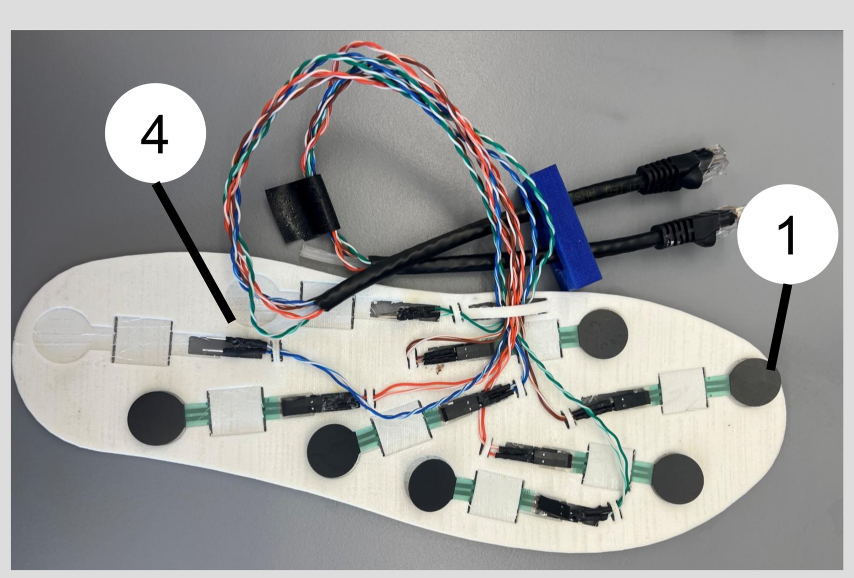
- 3D printed on a Bambu H2D, with a pause added in slicing for electronic integration
- FSR's calibrated by mapping compression testing results to FSR voltage output
- Serviceable, "snap on" cover version with electronics integrated post-printing

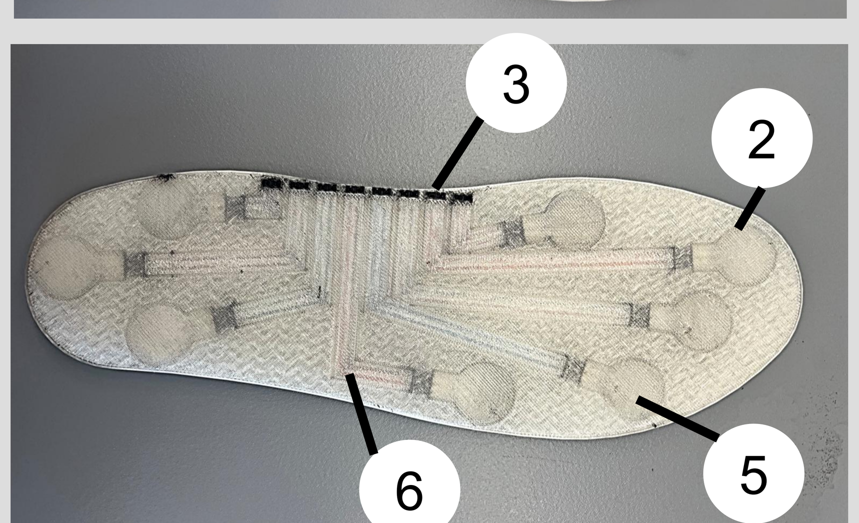
Design Choices:

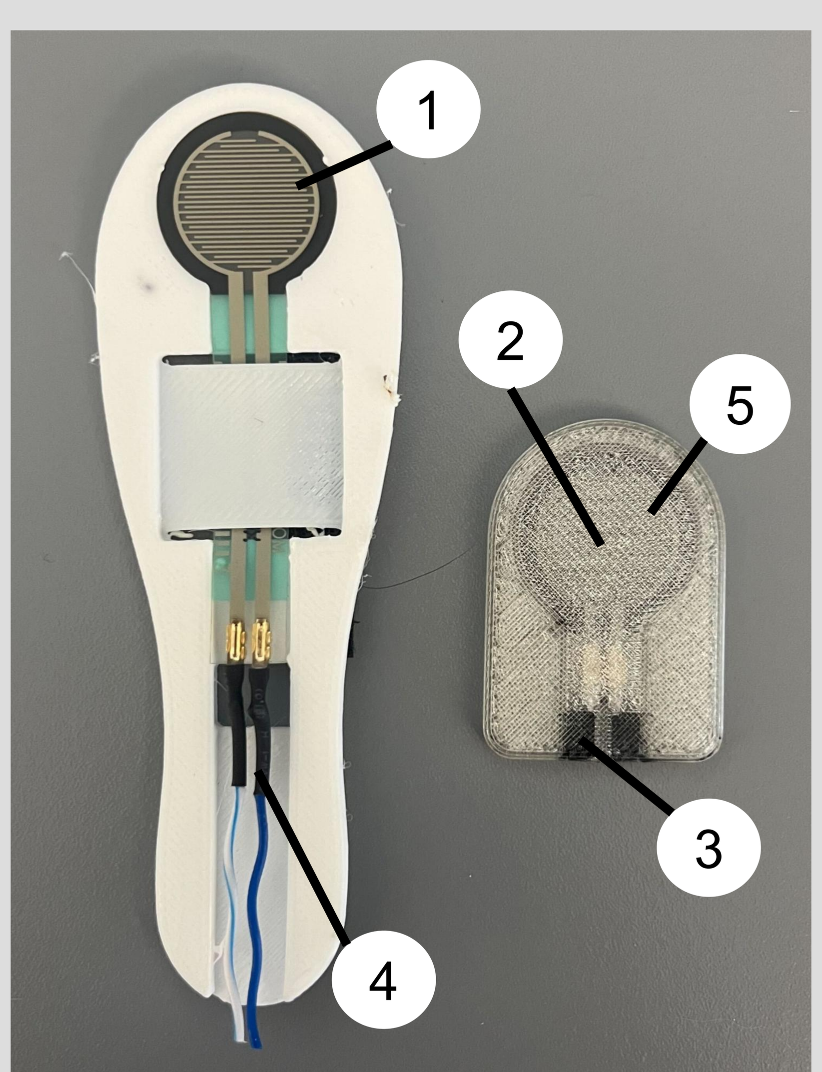
- Wire Clips: Tested two methods and the "clip" (top) was easier to use

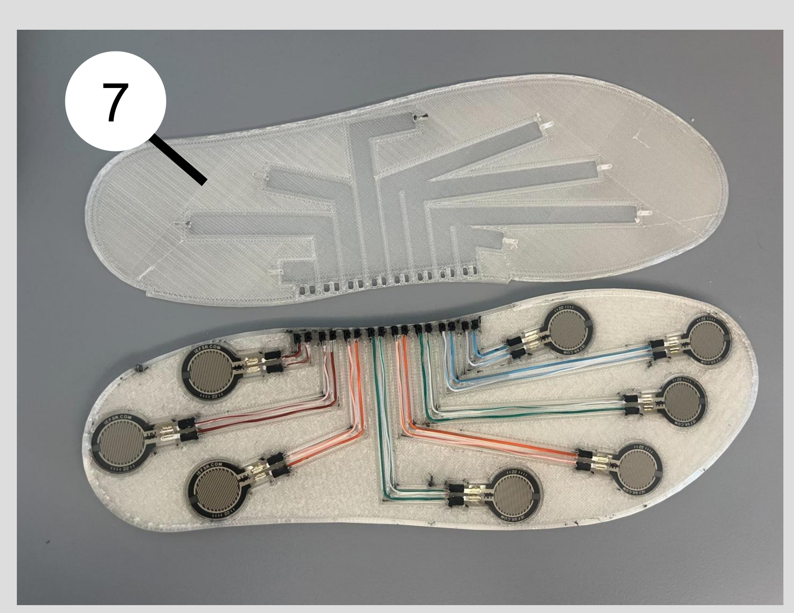


- Conductive Filament Selections: Eel Ninja Tek (95A) or Reprapper (90A)
  - Resistivity of 190.61 mm$^3$: Eel was 3.8 kΩ and Reprapper was 1.6MΩ









1. Long Tail FSR
2. Short Tail FSR
3. Conductive TPU Interface
4. Solder Connections
5. Enclosed FSRs
6. Internal Wires
7. Snap on cover

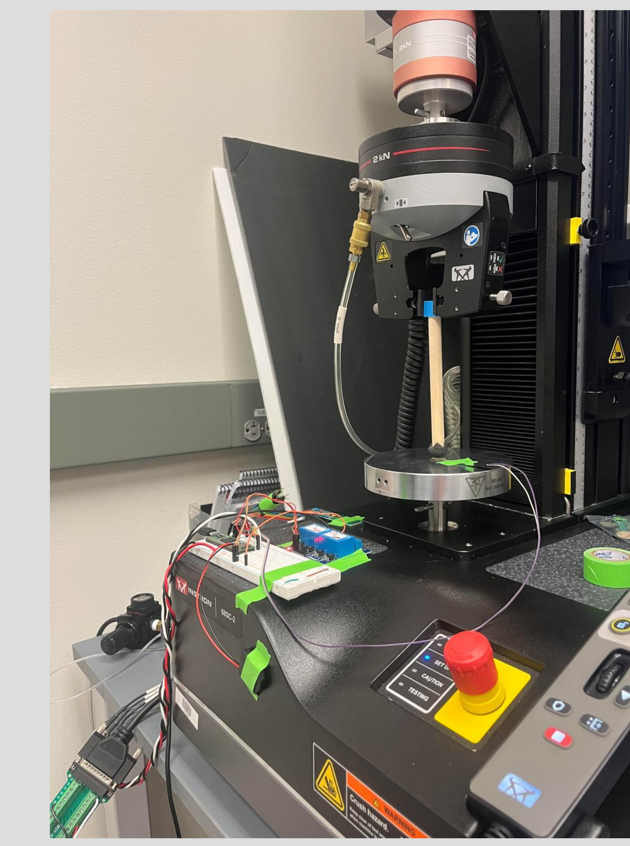
## Experiments and Results

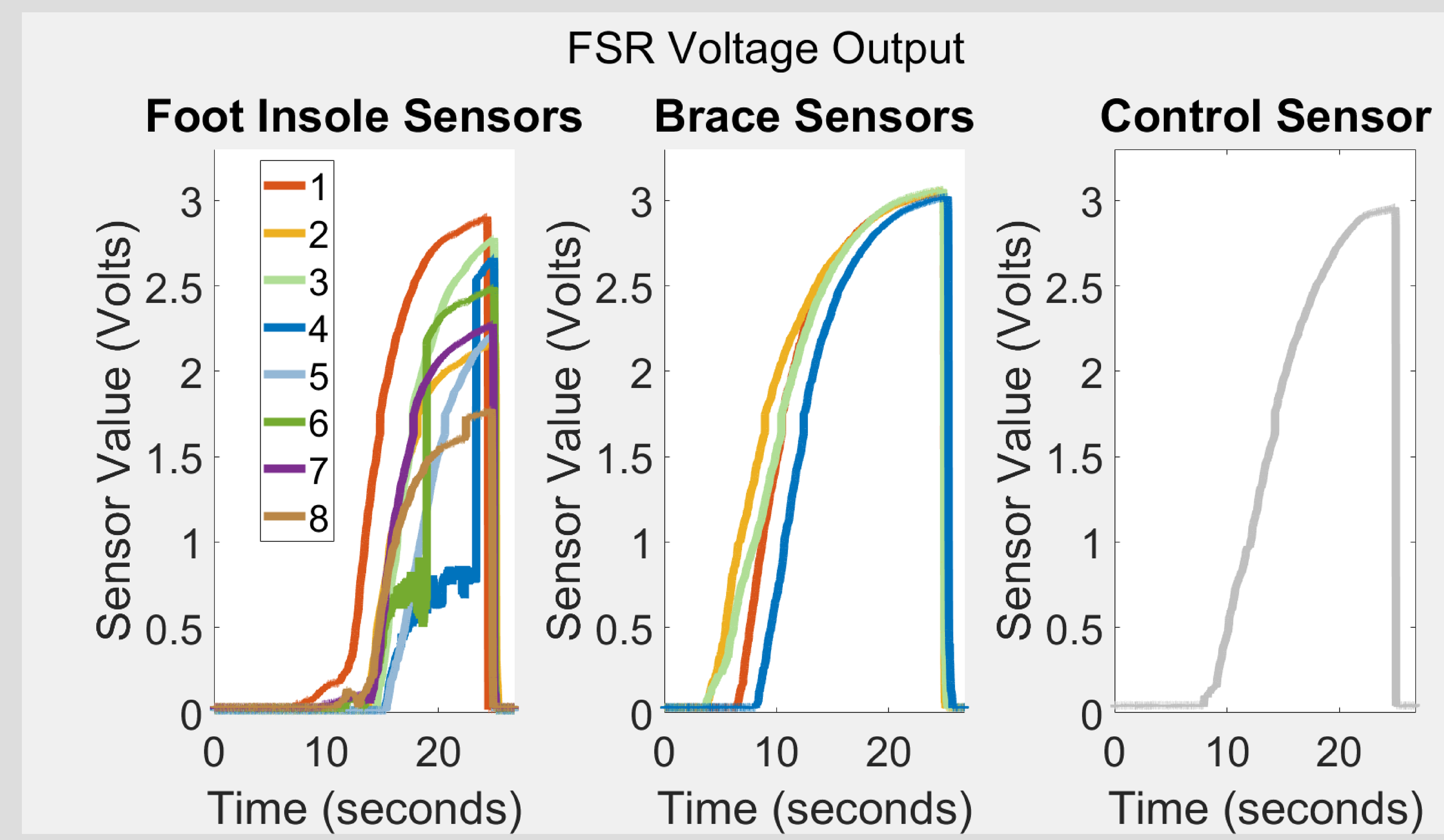
Experimental Methods:

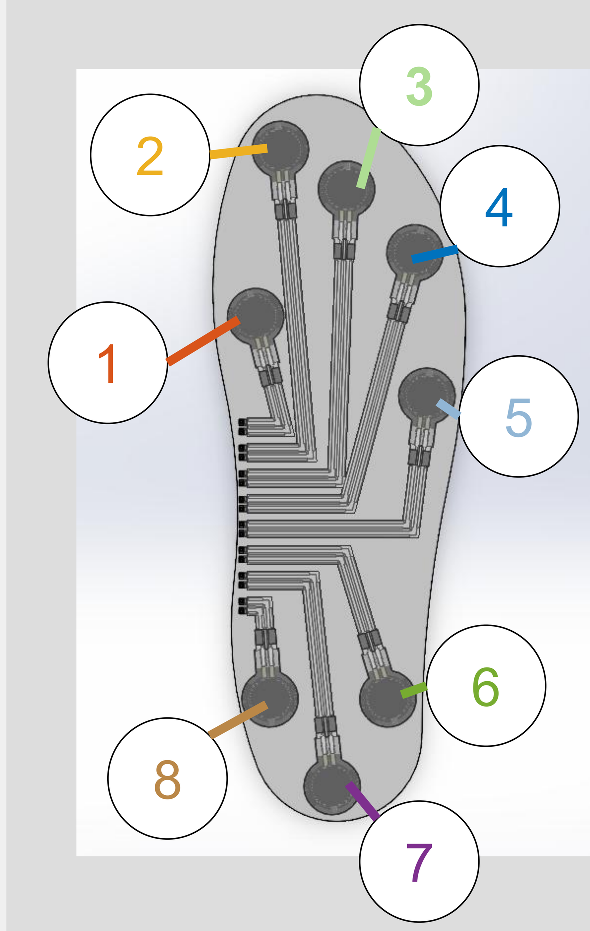
- To test the voltage output range of these sensors, they were linked to an Instron universal testing machine via a Teensy microcontroller with a 10kΩ voltage divider, and the FSR was compressed until the load cell read 200 N on the machine.

Results:

- The FSR's imbedded in the scoliosis brace sensor had the same voltage output as the control.
- Some of the FSR's imbedded in the foot insole reached the control level, but not all of them did.

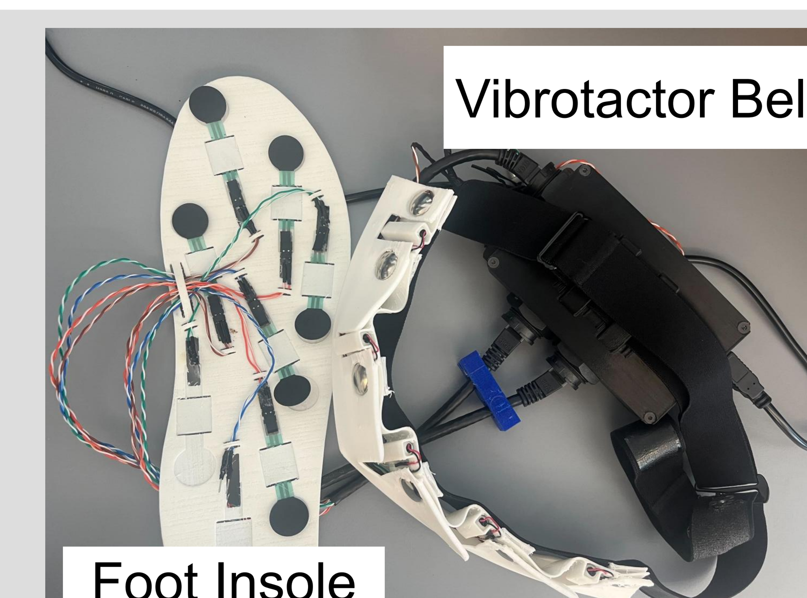




## Future Work

- Referred haptics or visual feedback for sensory augmentation from force sensor data
- Clinician training for prosthetic attachment with visual feedback or haptic feedback
- Device design validation by testing for sensor voltage output degradation after both short and long-term use

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